



# Dental Considerations In Respiratory Diseases

## Guidelines and Patient Clinical Management Strategies

BY

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# Introduction; The Bi-Directional Relationship

## ❑ **The Oral-Systemic Link:**

- **The mouth is the portal to the respiratory system.**
- In 2026, clinical focus remains on the "oral-lung axis," emphasizing how oral health directly impacts respiratory stability.

## ❑ **Risk Factors:** Many respiratory diseases share risk factors with oral diseases (e.g., smoking).

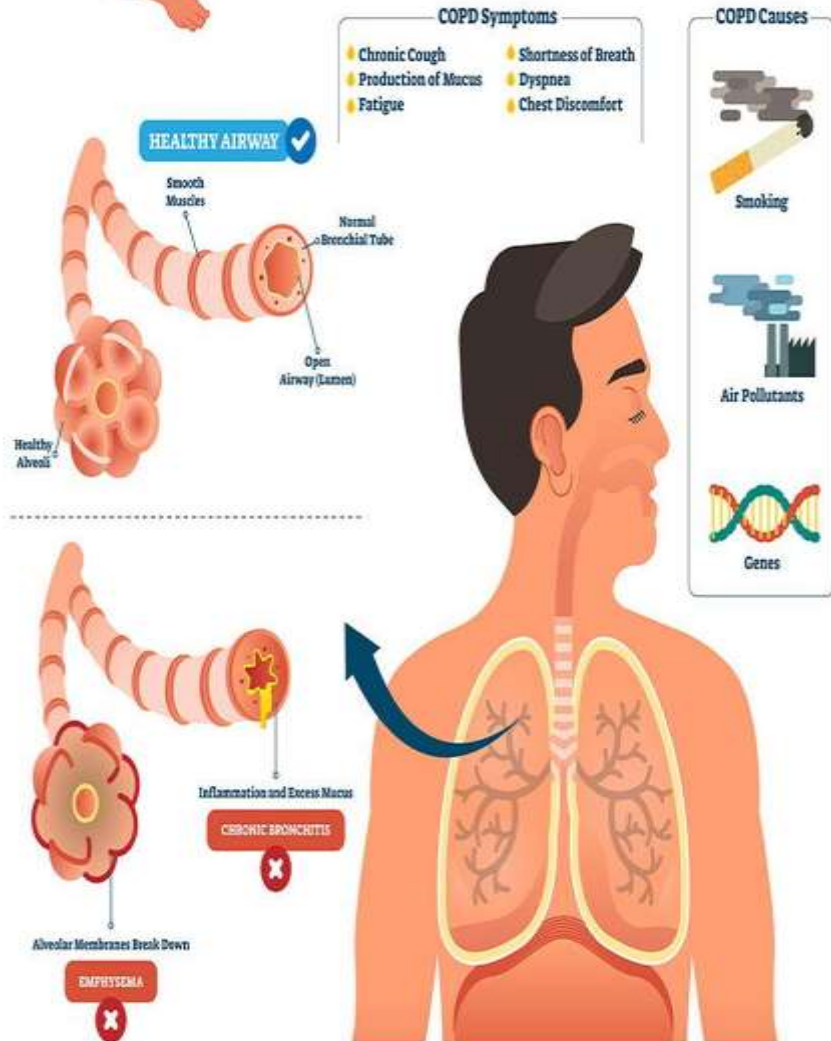
## ❑ **Primary Goal:** To provide safe dental care while minimizing the risk of acute respiratory distress or cross-contamination.

## ❑ **The link between the oral cavity and the lungs is characterized by three main pathways:**

- **Aspiration Pathway:** Inhaling oral bacteria (from plaque or infected gums) directly into the lower respiratory tract.
- **Inflammatory Pathway:** Systemic "distress signals" (cytokines like IL-6) travel from inflamed gums to the lungs, worsening chronic airway inflammation.
- **Medicative Pathway:** Drugs used to treat the lungs (inhalers, steroids) create secondary oral diseases.

# Major Respiratory Conditions in Dentistry

- Dental management typically focuses on these four categories:
  - **Asthma:** Reversible airway obstruction.
  - **COPD (Chronic Obstructive Pulmonary Disease):** Chronic bronchitis and emphysema.
  - Infectious bacterial disease; as **Tuberculosis (TB), Aspiration pneumonia;**
  - **Obstructive Sleep Apnea (OSA):** Airway collapse during sleep



## WHAT IS CHRONIC OBSTRUCTIVE PULMONARY DISEASE, (COPD) HOW DOES IT AFFECT ORAL THERAPY?

- Chronic obstructive pulmonary disease, abbreviated as COPD, is a form of chronic lung disease. COPD is an inflammatory lung disease that leads to airflow obstruction in the lungs. This makes it difficult for the individual to breathe adequately.
- The leading cause of COPD has been identified **as smoking**. However, COPD can also affect non-smokers who are exposed to smoke, such as in cases **of passive smoking**.
- Other irritants that can cause COPD are long-term exposure to fumes from burning fuel, such as wood, chemical fumes, air pollution, and living in poorly ventilated homes.

### The most common symptoms of COPD include:

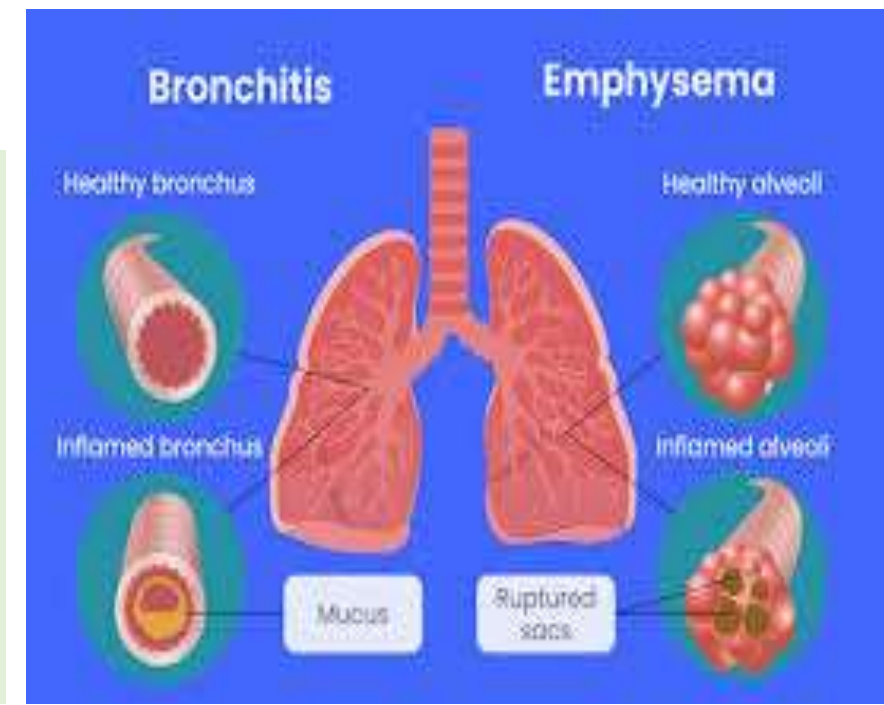
- Shortness of breath.
- Wheezing.
- Tightness in the chest.
- Chronic cough, which may or may not be associated with sputum production.
- Increased frequency of respiratory infections and colds



# TYPES OF COPD

The two most common conditions that contribute to COPD are Emphysema and Chronic Bronchitis.

- **Chronic Bronchitis** ; Inflammation and irritation of the bronchial tubes (airways) leading to increased mucus production.
- **Key Symptom:** A chronic productive cough (coughing up phlegm) for at least three months a year for two consecutive years.
- **Emphysema** is a condition in which the alveoli (small air sacs in the lungs) are destroyed as a result of exposure to chronic smoke or irritating gases and particulate matter.



## CHRONIC BRONCHITIS

CLINICAL DIAGNOSIS: DAILY PRODUCTIVE COUGH FOR THREE MONTHS OR MORE, IN AT LEAST TWO CONSECUTIVE YEARS

OVERWEIGHT AND CYANOTIC



ELEVATED HEMOGLOBIN



PERIPHERAL EDEMA

RHONCHI AND WHEEZING



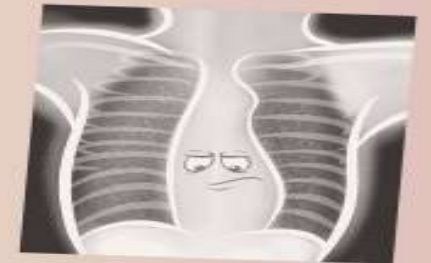
## EMPHYSEMA

PATHOLOGIC DIAGNOSIS: PERMANENT ENLARGEMENT AND DESTRUCTION OF AIRSPACES DISTAL TO THE TERMINAL BRONCHIOLE

OLDER AND THIN

SEVERE DYSPNEA

QUIET CHEST



X-RAY: HYPERINFLATION WITH FLATTENED DIAPHRAGMS

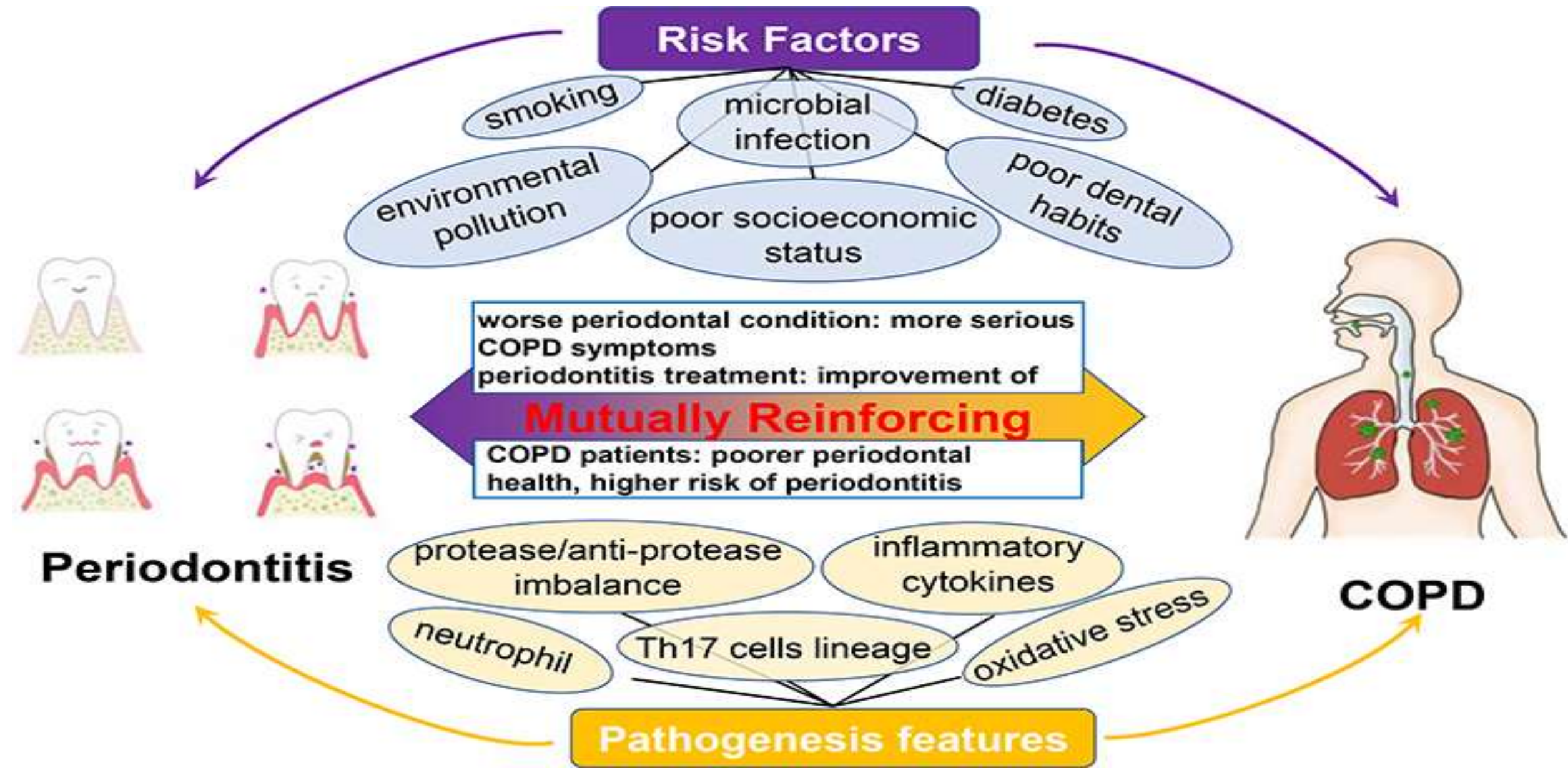


# Clinical Considerations IN COPD

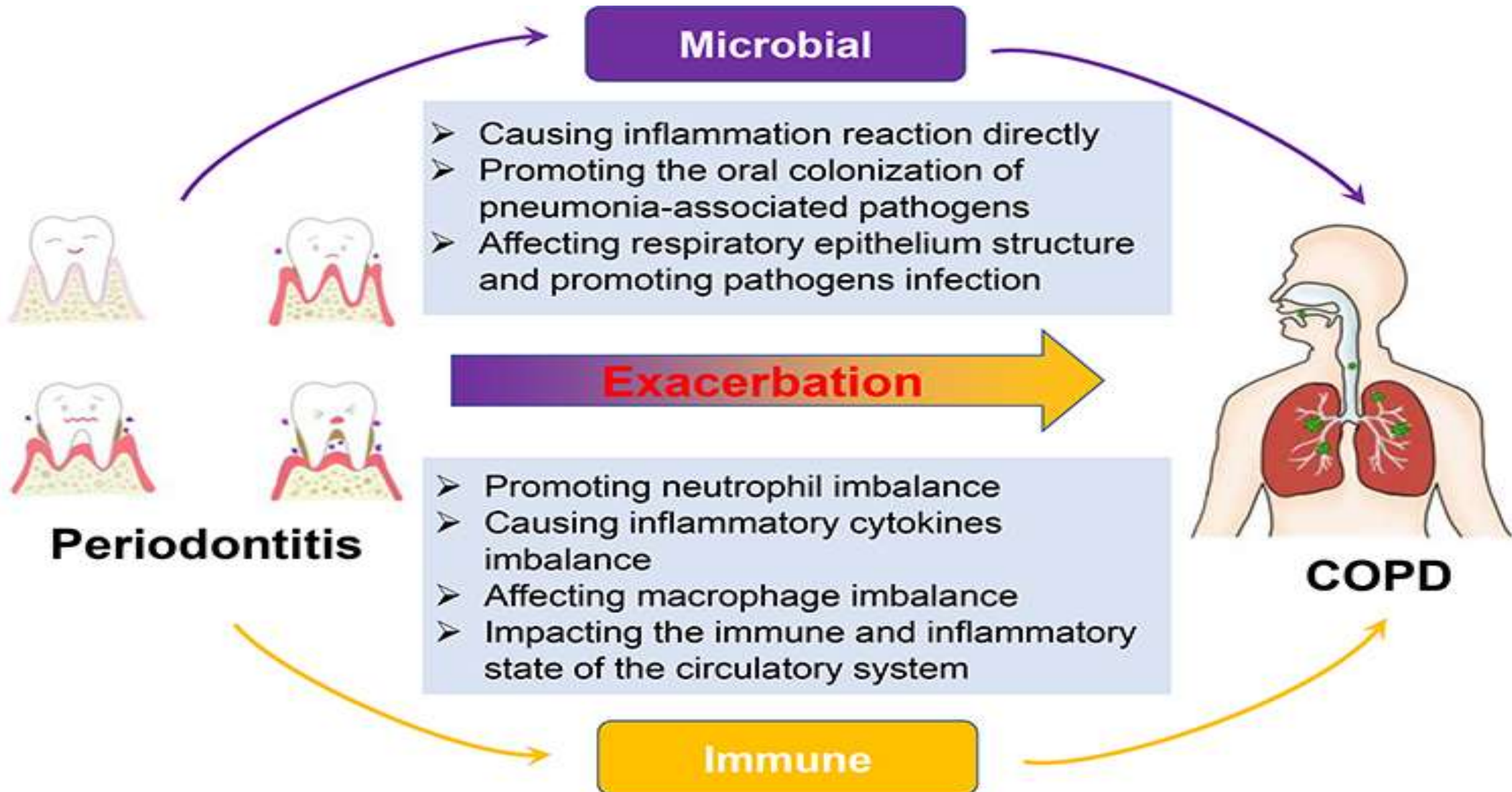
## ❑ Dental Management in COPD:

- **The Hypoxic Drive:** In advanced COPD, the brain relies on low oxygen levels (rather than high CO<sub>2</sub>) to stimulate breathing.
- Administering high-flow oxygen can lead to **respiratory arrest**. Limit supplemental O<sub>2</sub> to 2–3 L/min.
- **Avoid Nitrous Oxide (N<sub>2</sub>O):** It can accumulate in enlarged air spaces (bullae) in the lungs, potentially causing a **pneumothorax**.
- **The "Upright Rule":** Patients often have "orthopnea." If reclined, they may experience the sensation of drowning due to the weight of abdominal contents on the diaphragm
- **Patient Positioning:** Avoid the prone (flat) position; treat the patient in a **semi-upright** position to assist breathing.
- **Stress Reduction:** Keep appointments short and maximize morning scheduling.
- **Rubber Dam Use:** Use with caution as it may create a feeling of suffocation





The association between periodontitis and COPD. Periodontitis and COPD share some associations through various aspects, including epidemiological and clinical phenotypes, risk factors and pathogenesis features.



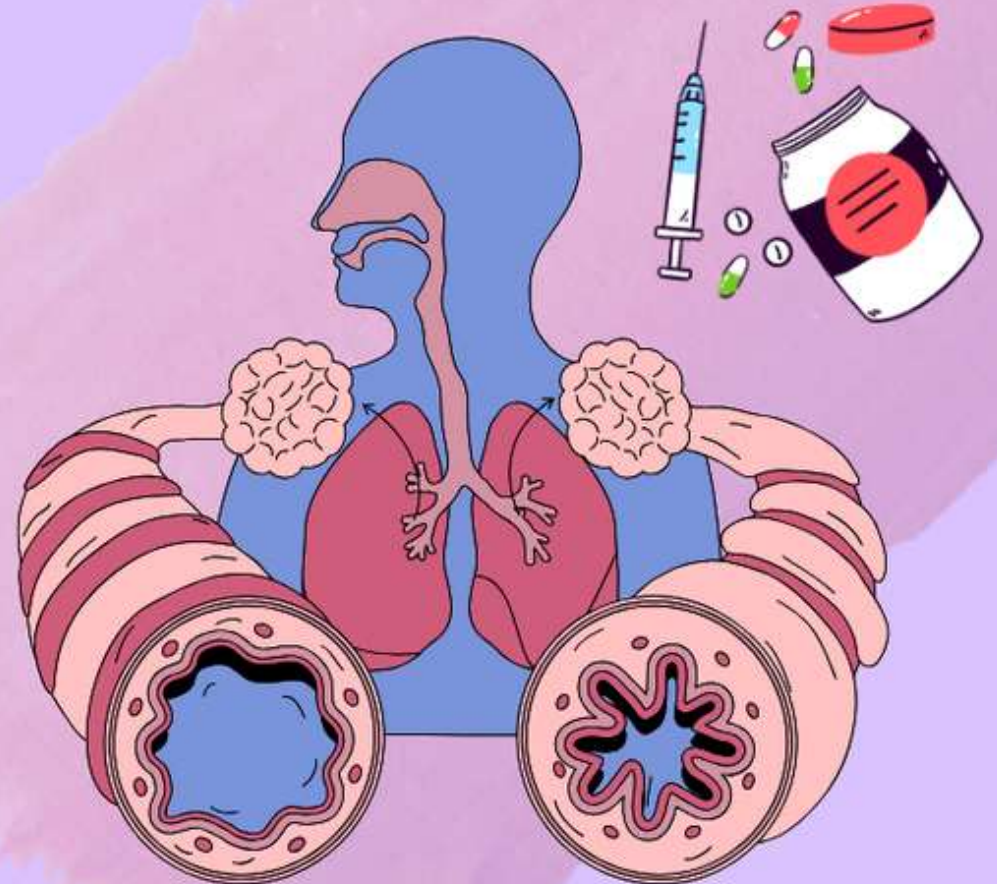
The potential mechanisms by which periodontitis affected COPD. Periodontitis promotes COPD progression mainly through causing imbalances in the microbial community and immune system.



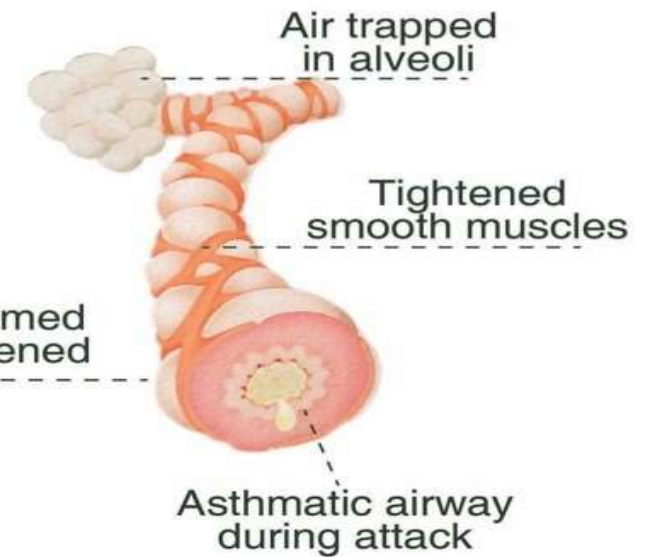
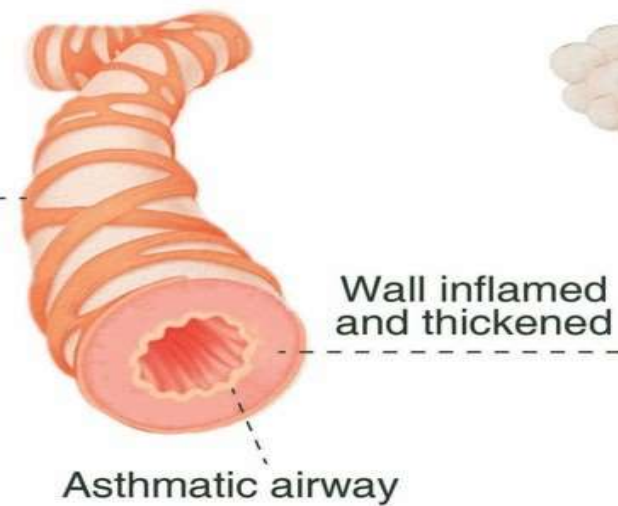
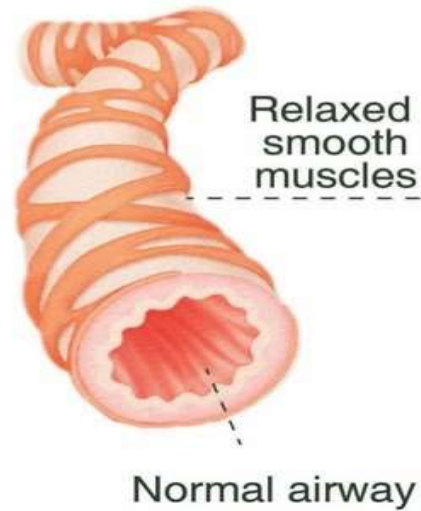
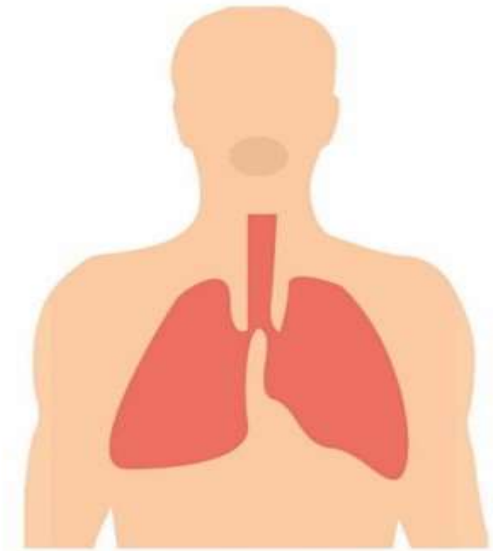
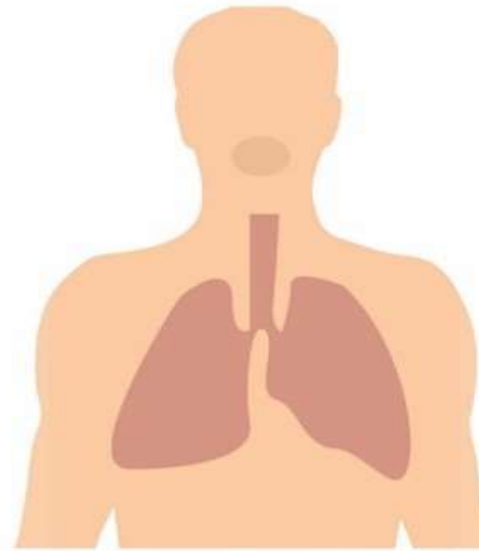
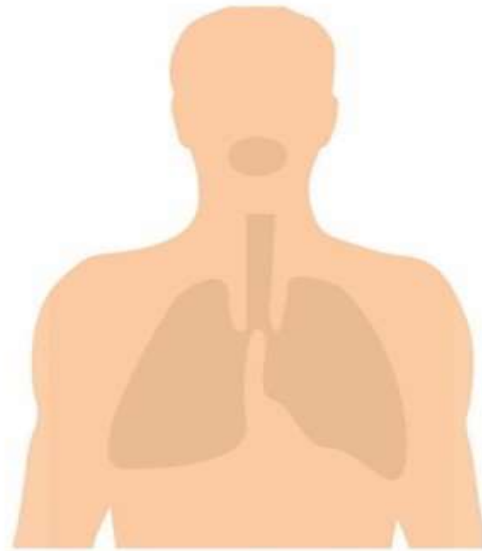
# BRONCHIAL ASTHMA

- Asthma is a common chronic disease worldwide
- Characterized by **chronic airway inflammation**.
- Causing symptoms such as wheezing, shortness of breath and cough.
- That varies in frequency and intensity together with variable expiratory airflow limitation.

## ASTHMA MEDICATIONS & DENTAL DRUG INTERACTIONS



# ASTHMA and YOUR AIRWAYS



**Intrinsic factors:**

Non-allergic/non-atopic  
Endogenous  
Emotional stress  
GERD  
Vagally mediated response

**Extrinsic factors:**

Allergic/atopic  
Most common  
IgE-mediated  
Pollen, dust, house mites, animal dander,  
odorants

**Asthma  
Triggers****Drug-induced:**

Aspirin & NSAIDS  
B-blockers & ACE inhibitors  
Metabisulfite preservatives  
Nuts, shellfish, strawberry

**Exercise-induced:**

Physical activity  
Thermal changes  
Inhalation during cold weather





Allergens



Work



Respiratory Infections

## Inflammatory Factors

## TRIGGERS for an ASTHMA ATTACK



Strong odours



Cold air and temperature change



Exercise



Stress and emotions



## Irritants



Pollutants



Food additives



Gastric reflux



Tobacco



Medication

## Others

## Asthma Investigations

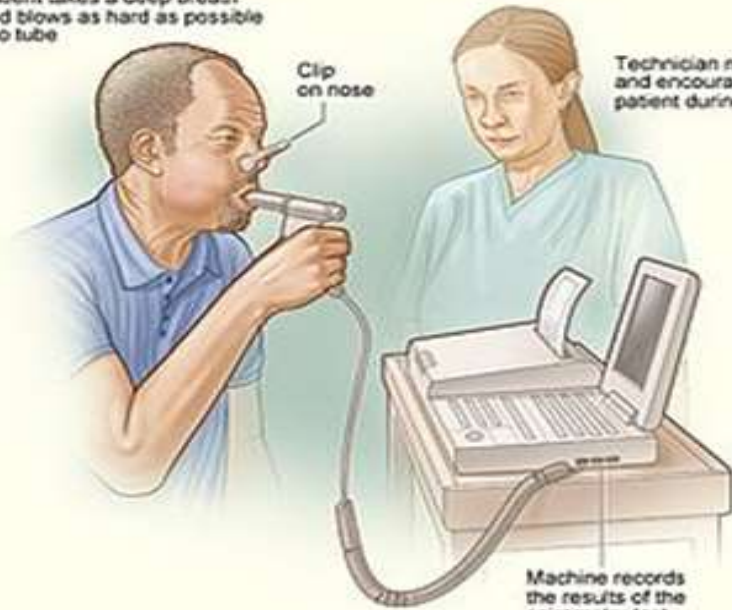
- spirometry
- peak flow meter
- radiology
- allergy skin test

Patient takes a deep breath and blows as hard as possible into tube

Clip on nose

Technician monitors and encourages patient during test

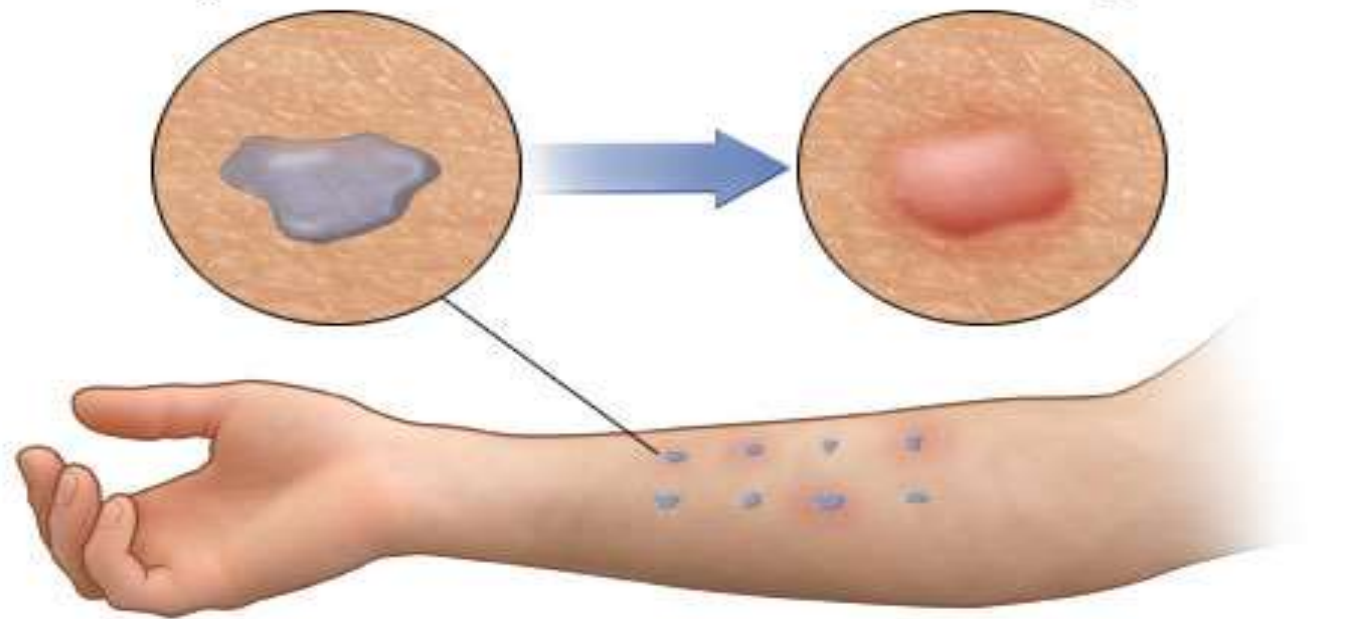
Machine records the results of the spirometry test



## DIAGNOSING ASTHMA

Allergen solution is placed on skin

Positive test: Skin is red and itchy



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# Clinical Considerations

- **The Sulfite Factor:** 5–10% of severe asthmatics are sensitive to **metabisulfite preservatives** found in local anesthetics containing epinephrine. For these patients, use plain anesthetics (e.g., **3% Mepivacaine**).
- **Drug-Induced Attacks:** Avoid NSAIDs and Aspirin. They can shift arachidonic acid metabolism toward the leukotriene pathway, triggering a severe "aspirin-induced" asthma attack.
- **Oral Manifestations:** \* **Xerostomia:**  $\beta_2$ -agonists reduce salivary flow, altering the oral microbiome.
- **Palatal Erosion:** GERD is common in asthmatics (due to theophylline or  $\beta$ -agonists relaxing the esophageal sphincter), leading to palatal enamel erosion.
- **Candidiasis:** Chronic use of inhaled corticosteroids (ICS).
-



# Dental Management: Asthma

- **Pre-treatment:** Ensure the patient has their "rescue" inhaler (albuterol) on the bracket table.
- **Trigger Avoidance:** Reduce anxiety; avoid strong odors or aerosols if they trigger the patient.
- **Medication Side Effects:** \* **Xerostomia:** Inhalers can reduce salivary flow.
  - **Candidiasis:** Steroid inhalers increase the risk of oral thrush (advise rinsing after use).
- **Contraindications:** Avoid aspirin and NSAIDs if the patient has "Aspirin-Triad" asthma.

# Dental considerations

## Before Treatment

- **Asymptomatic & Controlled:** Only treat patients with well-controlled asthma; reschedule if symptomatic (coughing, wheezing).
- **Medication Review:** Note use of inhalers (especially steroids, beta-agonists) and carry reliever inhaler to appointments.
- **Anxiety Management:** Reduce stress; consider mild anxiolytics if needed, as anxiety can trigger attacks.
- **Emergency Prep:** Have supplemental oxygen and epinephrine ready.

# Dental considerations

## During Treatment

- Airway Management:** Use a rubber dam to prevent inhalation of debris, keep patient comfortable, and monitor breathing.
- Avoid Triggers:** Be cautious with dental materials like methyl methacrylate, certain sealants, and even cotton rolls, which can trigger asthma.
- Anesthetic Choice:** Use bronchodilatory agents (e.g., sevoflurane) and avoid those that increase airway resistance.
- Stop if Needed:** Immediately stop if coughing, wheezing, or shortness of breath occurs; administer usual treatment.



# Dental considerations

## Oral Health Risks & Prevention

- **Dry Mouth (Xerostomia)**: Mouth breathing and beta-agonists reduce saliva, increasing decay/erosion risk; use fluoride, Xylitol, and proper hygiene.
- **Candidiasis (Thrush)**: Steroid inhalers alter oral flora; good hygiene and antifungal rinses can help.
- **Increased Caries**: Higher risk from dry mouth and altered pH; prescribe fluoride rinses/gels.
- **Gum Disease**: Increased risk due to mouth breathing and medication side effects.

# Acute Asthma During Dental Procedure

## Patients with *acute severe asthma* typically have:

1. Inability to complete a sentence in one breath
2. Respiratory rate  $\geq 30$  breaths per minute
3. Tachycardia  $\geq 120$  beats/min.
4. O<sub>2</sub> saturation  $< 90\%$

## How to manage

- **Don't Be Afraid**
- Discontinue the dental procedure and allow the patient to assume a comfortable position.
- Establish and maintain a patent airway and administer  **$\beta 2$  agonists** via inhaler or nebulizer.
- Administer **oxygen 6** liters via face mask, nasal cannula (oxygen saturation levels should be 93% to 95% in adolescents)
- If no improvement is observed and symptoms are worsening, administer epinephrine subcutaneously

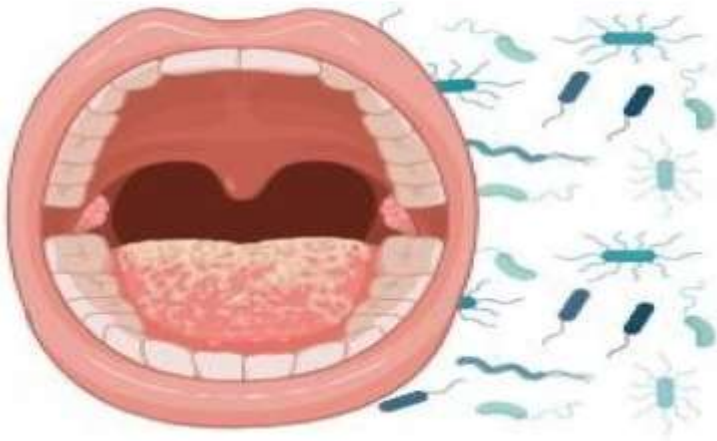
# Dental Surgery And Asthma

- **Sedation** → Hydroxyzine and benzodiazepines recommended.
- Avoid narcotics and **barbiturates** due to their histamine releasing properties → bronchospasm and potentiated allergic response.
- **Intravenous sedation** → Use extreme caution due to limited control of the airway.
- Avoid general anesthesia if possible **Enflurans or isoflurane** are preferred to **halothane**
- **Nitrous oxide with oxygen** is preferred to intravenous sedation in mild to moderate asthmatics after medical consultation.
- **Local anesthesia containing epinephrine or levonordefrin** also contain sulfite component which may precipitate asthmatic attack.

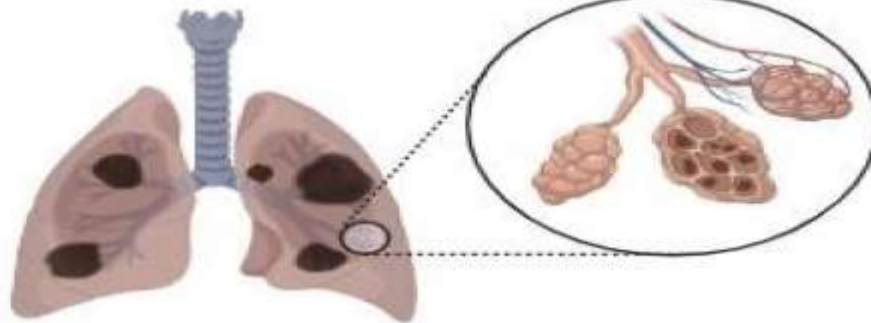
# Oral microbiome disorders (dysbiosis)

- This occur when the balance of mouth bacteria is disrupted, leading to increased pathogenic bacteria and reduced beneficial microbes. Common conditions include periodontitis, dental caries, and bad breath (halitosis)

**Oral Microbiom Disorder leads to:**



**Chronic  
Obstructive  
Pulmonary  
Disease**

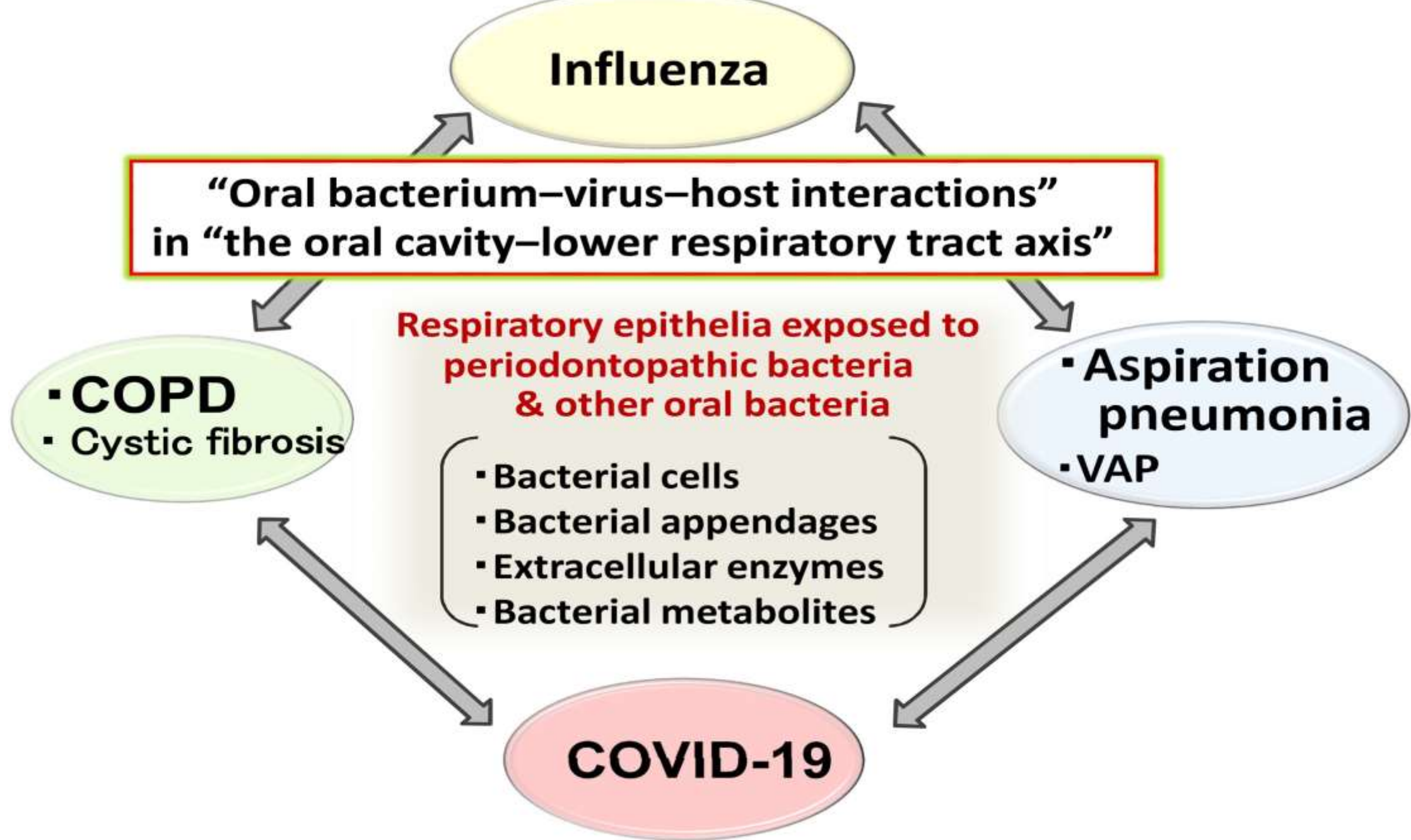


**Exacerbations of  
Asthma**



**Pneumonia**



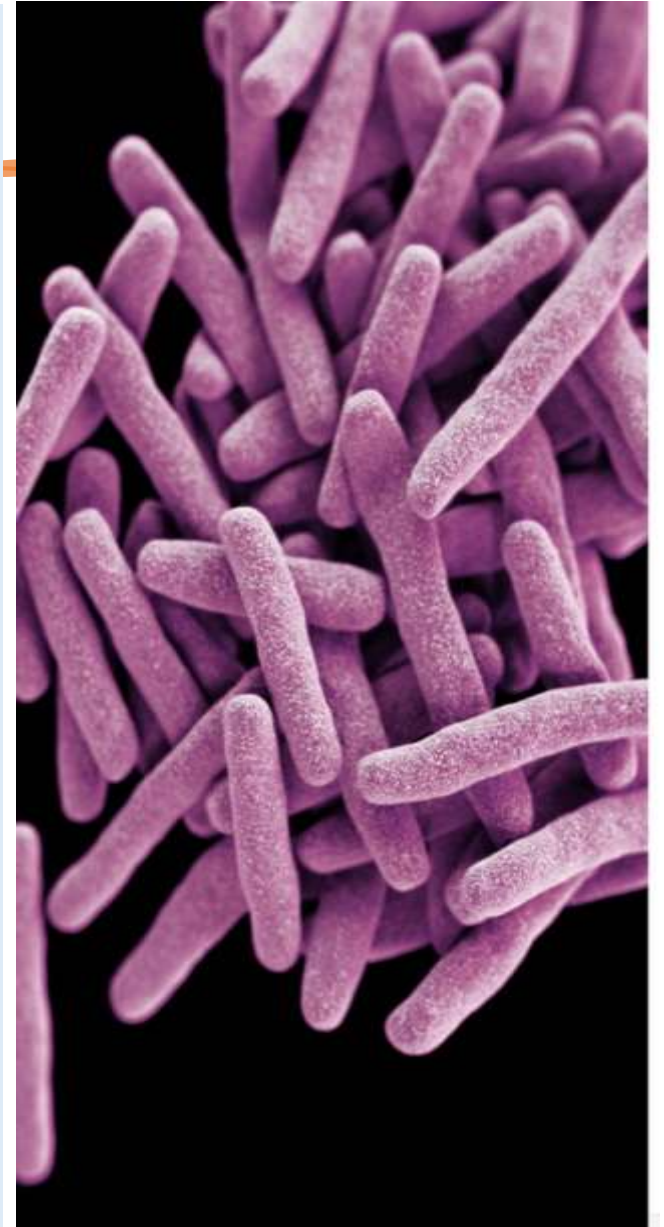


# Dental Considerations in Pneumonia

- **Oral-Lung Axis:** Poor oral hygiene allows bacteria to multiply in the mouth and potentially travel to the lungs, causing pneumonia.
- **Prevention:** Regular professional cleaning by a dentist is critical for reducing the risk of pneumonia, especially in older adults, with studies showing a reduction in recurrence by nearly 50%.
- **Appointment Management:** Patients with active, contagious pneumonia should not attend dental appointments to prevent spreading the infection.
- **Procedure Safety:** For patients with compromised breathing, chair positioning should be adjusted, and the use of rubber dams may be necessary.

# Types of Tuberculosis

- Tuberculosis (TB) is an infectious respiratory disease caused by the bacteria called **Mycobacterium Tuberculosis**. It usually infects the lungs and may also spread to other parts of the body like your brain or spine.
- Mycobacterium Tuberculosis gets transmitted from person to person through tiny droplets released into the air via coughs and sneezes.
- Tuberculosis can be fatal if not treated properly.
- Tuberculosis is classified into two types:
- **Latent TB:** In this type of infection bacteria is present in the body but in an inactive form and cause no symptoms. It is also known as inactive TB. Though it is in an inactive state it can become active and thus require proper treatment.
- **Active TB:** In this type of infection bacteria present in the body multiply and make the person sick. A person suffering from active TB can spread the disease to others.
- Sometimes active TB and latent TB may become drug resistant which means that certain medications don't work against them.





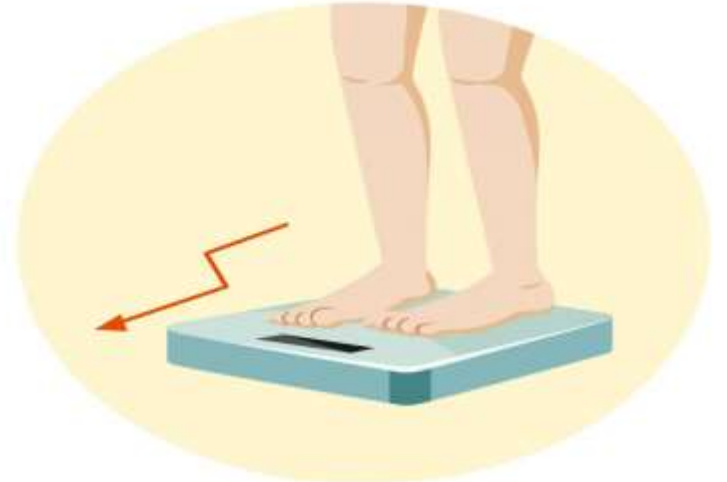
# Tuberculosis Symptoms



**fever**



**fatigue**



**weight loss**



**persistent cough**



**blood in cough**



**night sweats**





*Oral tuberculosis  
lesion*



*Tuberculosis lesion on tip of  
the tongue.*

# Dental Considerations in Tuberculosis (TB)

- All persons in the dental office who have the potential for exposure to *M. tuberculosis* can be tested for infection, either by whole-blood **interferon gamma** release assay, or a two-step baseline **tuberculin skin** test **at the beginning of employment**
- **Infection Control (Active TB):** Patients with active pulmonary TB should not receive routine dental treatment, as TB is spread via airborne droplets. Non-urgent care must be postponed until the patient is no longer infectious (usually after 2–3 weeks of drug therapy and a negative sputum report).
- Continue isolation until sputum smears **are negative for 3 consecutive** determinations (usually after approximately **2-4 wks.** of treatment).
- **Standard surgical face masks are not adequate** to protect against tuberculosis transmission.
- **Emergency Care for Active TB:** If emergency treatment is necessary, it must be provided in a facility with proper infection control, such as a negative-pressure room.
- **Preventive Measures:** Dental personnel should wear specialized masks (N95/FFP3) rather than standard surgical masks, which do not protect against TB.
- **Latent TB:** Patients with latent TB (no symptoms) can be treated using standard precautions.
- **Oral Manifestations:** Oral lesions of TB, often appearing as painful, chronic, irregular ulcers (commonly on the tongue or gingiva), should be included in differential diagnoses for suspicious lesions.
- **Drug Interactions:** Antitubercular drugs (e.g., rifampicin) can cause side effects like thrombocytopenia, hepatotoxicity, and increased cortisol metabolism.

# Dental considerations for patients with pneumonia and tuberculosis

- Dental considerations for patients with pneumonia and tuberculosis (TB) focus on preventing the transmission of infectious agents, managing oral manifestations, and adapting treatment based on the severity of the illness.
- **Summary of Key Management Strategies**
- **Screening:** Dentists should screen for TB symptoms, such as chronic cough, fever, and weight loss.
- **Postponement:** Defer elective treatment for active TB and contagious pneumonia.
- **Isolation/Environment:** Use high-volume suction and isolate patients with active infection.
- **Collaboration:** Liaise with the patient's physician before treatment.

# **Sleep Apnea & Dental Health**

 Sleep Care online



# Understanding the Link Between Sleep Apnea and Dental Health

## Why Your Dentist Might Be the First to Save Your Sleep

- It might seem surprising that your dentist could be the first person to spot a sleep disorder, but the mouth often displays the first warning signs **of Obstructive Sleep Apnea (OSA)**.
- When your breathing is interrupted during sleep, your body reacts in ways that can cause significant wear and tear on your teeth and gums.
- **What is Sleep Apnea? (The Basics)**
- **Definition:** A serious sleep disorder where breathing repeatedly stops and starts.
- **Obstructive Sleep Apnea (OSA):** The most common form, caused by throat muscles relaxing and blocking the airway.
- **The Connection:** Since the airway starts in the mouth and throat, **dentists are often the "first responders" in identifying symptoms.**
- **Core Impact:** Reduced oxygen levels and fragmented sleep.

# Common Dental Problems Linked to Sleep Apnea

- **Bruxism (Teeth Grinding/Clenching)**: Causes tooth wear, enamel loss, fractures, and jaw pain.
- **Dry Mouth (Xerostomia)**: Mouth breathing leads to reduced saliva, increasing risk for tooth decay (caries) and gum disease.
- **TMJ Disorders**: Jaw strain from clenching can cause temporomandibular joint pain and dysfunction.
- **Gum Disease**: Inflammation from OSA is linked to increased periodontal issues.
- **Scalloped Tongue**; If your tongue is too large for your mouth (a common physical cause of OSA), it may press against your teeth, creating "ridges" or a scalloped appearance along the edges.
- **Palatal Changes**: Alterations to the roof of the mouth can occur.

- **Red Flag #1: Bruxism (Teeth Grinding)**

- **The "Why":** When the airway collapses, the body subconsciously clenches the jaw to tighten airway muscles.
- **Dental Indicators:**
  - Worn-down, flattened, or chipped teeth.
  - Loss of tooth enamel and increased sensitivity.
  - Broken fillings or crowns.
- **Key Fact:** Patients with OSA are significantly more likely to experience nocturnal grinding.

- **Red Flag #2: Dry Mouth & Gum Disease**

- **The Cause:** Sleep apnea often leads to chronic mouth breathing.
- **The Effect:**
  - **Reduced Saliva:** Saliva is necessary to neutralize acid and wash away bacteria.
  - **Bacterial Overgrowth:** Leads to higher rates of plaque, cavities, and gingivitis.
- **Symptom:** Waking up with a "cotton mouth" or persistent bad breath (halitosis).

# The Dentist's Role in Management

- **Screening:** Dentists use questionnaires (like Epworth Sleepiness Scale) and oral exams to identify risk factors.
- **Referral:** They refer patients for formal sleep studies (polysomnography).
- **Oral Appliance Therapy (OAT):** Dentists fabricate custom devices for mild-to-moderate OSA:
  - Mandibular Advancement Devices (MADs): Move the lower jaw forward to open the airway.
  - Tongue Retaining Devices (TRDs): Hold the tongue forward.
- **Preventive Care:** Address dry mouth with products, manage decay, and monitor for complications.



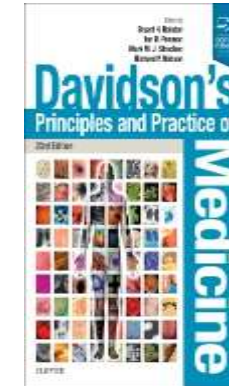
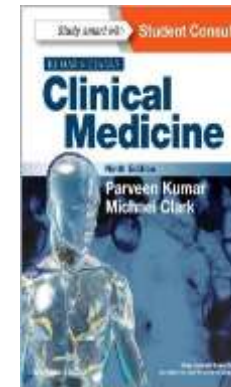
# Considerations for Oral Appliance Users

- **Side Effects:** Expect potential jaw discomfort (usually temporary), changes in saliva, and eventual bite/occlusal changes.
- **Monitoring:** Regular check-ups are crucial to adjust appliances, prevent bite changes, and ensure effectiveness.
- **Long-Term Care:** Oral appliance therapy requires lifelong dental monitoring
- **Important Patient Advice**
- **Inform Providers:** Always tell your dentist and medical team about your sleep apnea diagnosis before procedures, especially with sedation or anesthesia.
- **Multidisciplinary Care:** Optimal management involves collaboration between dentists, physicians, and sleep specialists.

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# References & Further Readings

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Thank you!









